Dapeng WANG (Phoenix)

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EDUCATION

THE HONG KONG POLYTECHNIC UNIVERSITY — Hong Kong, China
Bachelor of Engineering (Honors) in Mechanical Engineering — September 2020 – July 2024

- GPA: 3.80/4.3 (3.70/4.0 from WES calculation) (First Class Honour, Top 4%)
- Major GPA: 3.87/4.3
- Dean's Honours List 2021/22, 2022/23 (Top 10%)
- The Hong Kong Polytechnic University Scholarship 2023/24

RESEARCH EXPERIENCE

ROBOTIC FISH (CAPSTONE PROJECT) — Hong Kong, China
September 01, 2023 - May 13, 2024

- Design and develop a robotics fish under the supervision of Dr. Hui Tang
- The fish could successfully thrusting, diving & lifting.
- My jobs are dynamics modelling of lifting & diving and fish tail, building Kalman Filter for velocity and altitude control, signal processing and hardware design & manufacturing.
- This project received A+ as final grade (10% A+ rate).

STUDENT RESEARCH ASSISTANT IN RENEWABLE ENERGY ADVANCEMENT LAB — Hong Kong, China
Research Assistant — October 03, 2023 - January 31, 2024

- Conduct research on the continued work of PIML-DCNN model on quantify staggered-height configurations on convective heat transfer of PV arrays from URIS under the supervision of Dr. Mengying LI.
- Paper under review, 'Fast estimation of the convective cooling performance of Photovoltaics array with various configurations: a physics informed machine learning approach' to 'Enery & AI' as the first author.

- Helping the Lab apply for applying hyper computation platform of The Hong Kong Polytechnic University.
- Helping the Lab daily affairs, like help writing grants and reports.

UNDERGRADUATE RESEARCH AND INNOVATION SCHEME (URIS) — Hong Kong, China
Undergraduate researcher in scheme — June 07, 2022 – August 31, 2023

- Conduct research on the convective heat transfer effect of staggered-height arrangement on PV farms under the supervision of Dr. Mengying LI.
- Submitted a conference paper, "Efficiency improvement of solar Photovoltaic arrays using height-staggered configurations", to PolyU Research Student Conference (PRSC 2023), and the abstract has been accepted.
- Proposed a new method with Physics-Informed Machine Learning and Deep Convolution Neural Network combined model (PIML-DCNN) to solve the interpretability of PV arrays staggered-height configurations on convective heat transfer in quantitative view. Such method and relative experiment validation results has been recorded in URIS Final Report.
- The final report of URIS was rated 'very satisfactory' by the committee.

STUDENT RESEARCH ASSISTANT IN RENEWABLE ENERGY ADVANCEMENT LAB — Hong Kong, China
Research Assistant — Jan 24 - April 29, 2022

- Conduct research on adaptive cooling method combining radiative and convective heat transfer of photovoltaic panels under the supervision of Dr.LI Mengying.
- Building prototypes for flapping convective heat transfer of PVs and conducting experiment.

2021 SPRING CETUS-TALK ONLINE RESEARCH — Hong Kong, China
June 06 – Augst 26, 2021

- Conduct research about a typical autonomous driving navigation algorithm under the supervision of Prof. Fumiya Lida from Cambridge University and get the recommendation letter from him.
- Published the conference paper, "New Improvement of 'Dynamic Repulsive Potential Factor' on Artificial Potential Field Path Planning Algorithm" on 2021 2nd IYSF Academic Symposium on Artificial Intelligent and Computer Engineering.

EXTRA-CURRICULAR ACTIVITIES

MATHEMATICS FOR ENGINEERING ONLINE SUMMER PROGRAMME (ONLINE MODE)
Hong Kong, China

July 26 – July 30, 2021

- Study linear algebra, numerical analysis from Dr. Nick Godfrey from Girton College, Cambridge.

JOINING HONG KONG REGIONAL COMPETITION 2022 OF ENACTUS Hong Kong, China
13th President of Enactus HKPU & Project Manager May 24, 2021 – June 25, 2022

- Working as 13th president of Enactus HKPU and project manager of team 'Snakecup', preparing for the regional competition of year 2022.
- Developing a template app service for our project by studying Javascript and html by myself.
- Distributing workload of the project based on teammates' studying major and personalities.
- Successfully joining and finishing 2022 regional competition of Enactus HK.

2022 GLOBEX JULMESTER (ONLINE MODE) Beijing, China
Exchange student July 04 – July 22, 2022

- Attending 2022 Globex Julmester held by Peking University.
- Studying Intelligent Manufacturing and Service Systems from Professor Andrew KUSIAK, and Control Theory from Professor. Xun HUANG.
- Achieve high scores of classes: Intelligent Manufacturing and Service Systems: 97, Control Theory: 92.

PROOF-OF-CONCEPT (POC) FUNDING SCHEME 2022 Hong Kong, China
Start-up team member, CTO October 19 – November 28, 2022

- Working as the CTO of 'Snakecup' team continuously, and joined Proof-Of-Concept (POC) Funding Scheme 2022 in PolyU .
- Admitted to POC pith competition and won Dr. Winnie S M Tang- PolyU Student Innovation & Entrepreneurship Scholarship.

RS INNOVATORS CHAMPIONSHIP 2023 Hong Kong, China
Team leader & AI model designer of team EA Dynamics November 30, 2022 – April 05, 2023

- Lead team named EA Dynamics joined RS Innovators Championship 2023.
- Proposed a mode based on LSTM to provide new strategy of logistics arrangement based on logistics data, improving distribute center working efficiency.
- Lead the team become 10 finalist teams worldwide and joined the final round the worldwide competition.

COMMUNITY SERVICES

HABITAT GREEN IN EAST AFRICA Rwamagana District, Rwanda
Volunteer student helper and technician July 22 – August 06, 2023

- Going to Rwanda to build small solar power grid for power supply for villages which didn't have electrical power.
- Working as a technician to build the grid infrastructure by assisting wiring, circuit connecting, and training local youth to install power generation system.
- Interviewed by Hong Kong Ta Kung Wen Wei Media Group, HKTKWW. Relevant article was published on HKTWW and Rwanda local media, IGIHE.

SELECTED AWARDS

DR. WINNIE S M TANG- POLYU STUDENT INNOVATION & ENTREPRENEURSHIP SCHOLARSHIP

- 'Dr Winnie S M Tang-PolyU Student Innovation & Entrepreneurship Scholarship' is the funding support of PolyU Student Entrepreneurial Proof-of-Concept (POC) Funding Scheme since 2022. (POC) Funding Scheme is a funding initiative launched in 2017 as one of the key entrepreneurship education components at PolyU. POC Fund aims to instil a problem-driven innovation mindset into young talents through experiential learning and student-initiated research and development. This funding is awarded to the undergraduate or graduate team who built entrepreneurship project has significant potential of making impacts on Hong Kong society. Up to $20,000 for each awarded team, and we received &15,000. My project 'Snakecup' and her team is awarded for the potential great healthcare and sustainable development impacts on Hong Kong community from the project.

PROFESSOR LEUNG TIN-PUI MEMORIAL SCHOLARSHIP

- The selection is based on the overall merits in the fields of academic performance, extra-curricular activities as well as community services. This scholarship with $20,000 would be awarded to students with high GPA performance (3.5+),

Outstanding in extra-curricular activities and community services, and strong aspirations on engineering.

THE HONG KONG POLYTECHNIC UNIVERSITY SCHOLARSHIP 2023/24

- The selection is based on the overall merits in the fields of academic performance, extra-curricular activities as well as community services. This scholarship with $40,000 would be awarded to students with high yearly GPA performance (3.7+).